

Sobhan Salarian — Data Scientist

Sydney, NSW, Australia

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Profile

Data Scientist and Master of Data Science student with hands-on experience in Python and SQL for data cleaning, analysis, and pipeline support. Skilled in machine learning fundamentals, data visualization (Power BI, Matplotlib/Bokeh), and NLP workflows (Hugging Face, translation evaluation). Interested in building reliable analytics systems and translating data into actionable insights.

Education

Macquarie University

Master of Data Science

Sydney, NSW

2025–Present

Kermanshah University of Technology

Bachelor of Computer Engineering

Kermanshah, Iran

2018–2023

Adel High School of Isfahan University

High School (Mathematics Major)

Isfahan, Iran

2014–2018

Technical Skills

Programming: Python (Pandas, NumPy), SQL

ML/NLP: scikit-learn, Hugging Face (Transformers/Datasets), PyTorch (training workflows), model evaluation (BLEU/chrF)

Data Analysis: Data cleaning, EDA, statistical analysis, feature engineering

Big Data: MapReduce (MrJob), MongoDB (PyMongo), R-Tree indexing, skyline queries

Databases: MySQL, MongoDB

Visualization: Power BI, Matplotlib, Bokeh

Tools: Git, GitHub, Jupyter, VS Code

Web: HTML, CSS

Languages

Persian (Farsi): Native

English: Professional Working Proficiency

Professional Experience

IRISA Company

Data Science Intern

Isfahan, Iran

Jul–Sep 2022

- Cleaned and standardized datasets in Python by handling missing values, outliers, and inconsistent records to improve analysis readiness.
- Loaded curated data into MySQL using Python connectors and supported schema/query workflows for downstream reporting.
- Documented processing steps and assisted with data quality checks to improve pipeline reliability and data availability.

Projects

CRM Enhancement with AI-Powered Recommender System — AI, Project Management, Risk Analysis

- Designed a data-driven CRM solution using an **AI-based recommender system** to enhance customer engagement and personalization.
- Applied **Agile-Waterfall hybrid methodology** to manage project phases, deliverables, and milestones.
- Identified and mitigated key risks (**data quality, privacy, system integration**) using structured risk management strategies.
- Conducted **stakeholder analysis** and collaborated with cross-functional teams to align business requirements and ensure adoption.
- Defined **KPIs** targeting improvements in customer retention, conversion rates, and marketing ROI.

Bitcoin Price Prediction using Databricks — Machine Learning, Data Analysis ([Certificate link](#))

- Performed **exploratory data analysis (EDA)** and visualized Bitcoin market trends.
- Preprocessed datasets for **machine learning pipelines**.
- Implemented a **regression model** in Databricks to predict Bitcoin price movements.

Multilingual ED Triage Translation — NLP, Machine Translation, Hugging Face

2025

- Designed safe **Chinese↔English** translation for clinical **LEP** scenarios.
- Processed medical datasets and built **tokenization and validation pipelines**.
- Fine-tuned **NLLB-200** using **GPU training** with Hugging Face workflows.
- Evaluated models using **BLEU/chrF** and conducted error analysis.

Spatial Data Analysis — R-Trees, kNN, Skyline Queries

2025

- Implemented an **R-Tree index** for efficient spatial querying.
- Developed **kNN** and **skyline algorithms** for location-based analysis.
- Benchmarked **multiple algorithms** for performance and scalability.
- **Presentation:** Google Drive link

Big Data Weather Analysis — MapReduce, MongoDB

2025

- Processed large datasets using **MongoDB** and **MapReduce** with MrJob.
- Built **distributed data pipelines** for extreme weather detection and analytics.
- Documented **data workflows and system architecture**.

Tennis Data Analysis — Python, Pandas, Matplotlib

2024

- Analyzed datasets using **Pandas/NumPy** to identify performance patterns.
- Processed parquet data and created **reproducible analysis notebooks**.
- Visualized insights using **Matplotlib**.

Personal Portfolio Website — HTML, CSS, JavaScript

2024

- Built a responsive portfolio website using **HTML, CSS, JavaScript**.
- Deployed via **GitHub Pages** and managed version control with **Git**.

Flight Delay Analysis System — Python, Bokeh, Machine Learning

2023

- Analyzed delay patterns using **Pandas** and interactive visualizations in **Bokeh**.
- Developed a **recommendation model** for airline selection based on delay trends.
- Built **interactive dashboards** for data exploration.

Professional Certifications

Neo4j GraphAcademy

Neo4j & Generative AI Fundamentals

2025

- Covered **Generative AI fundamentals**, **LLMs**, and retrieval-augmented generation patterns (**RAG/GraphRAG**).
- Explored integrating **Neo4j** with generative AI applications and graph-based retrieval workflows.
- Credential: [Link](#)

Daneshkar Academy

Data Analysis Bootcamp

2024

- Covered statistics, SQL/MySQL, and data manipulation with **Pandas/NumPy**.
- Built dashboards and reports in **Power BI** and practiced Git/GitHub workflows.

Coursera

Python Programming for Everybody

2023–2024

- Completed Python fundamentals and intermediate concepts, including basic data analysis and database integration.